
Improved Automated Pneumonia Detection Using Deep Transfer Learning Enhanced with Attention Mechanisms in Pediatric Chest X-ray Images

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Abstract

Pneumonia remains a leading cause of mortality in children globally. Accurate and timely diagnosis via chest X-ray (CXR) is crucial but challenging due to subjective human assessment and limited specialist access. This paper proposes an enhanced automated pneumonia detection system, leveraging **deep transfer learning** and the **Xception** architecture. Our novel approach employs a **two-stage fine-tuning strategy** and an **enhanced classifier head** (with Dense layers, Batch Normalization, and Dropout). We address data imbalance with a **Focal Loss** function and **L2 Regularization**. Crucially, a **Convolutional Block Attention Module (CBAM)** is introduced to enable the model to dynamically focus on salient regions, improving diagnostic accuracy and interpretability. Experimental results show superior performance (accuracy 92.15%, AUC 0.9667) compared to previous reference models, offering a robust and effective solution for pediatric pneumonia detection.

1 Introduction

Pneumonia is a severe acute respiratory infection causing significant morbidity and mortality worldwide, particularly in children under five (1). Accurate and timely diagnosis is vital for effective treatment. Chest X-ray (CXR) is the primary diagnostic tool, but its interpretation is challenging due to required expertise, subjectivity, and limited access to specialists, often delaying diagnosis.

Deep learning, especially Convolutional Neural Networks (CNNs), has revolutionized medical image analysis, offering promising automated diagnostic assistance. However, applying deep learning to medical imaging, like pediatric CXR, faces obstacles such as data scarcity, class imbalance, and the need for accurate, interpretable models.

Our work addresses these challenges by developing an advanced automated pneumonia detection system. We leverage **deep transfer learning**, fine-tuning a pre-trained **Xception** model on a pediatric CXR dataset. Our main contributions are:

- A **two-stage fine-tuning strategy** optimizing model adaptability to CXR characteristics.
- An **enhanced classifier head** with multiple Dense layers, Batch Normalization, and Dropout for regularization.
- Integration of a **Focal Loss** function and **L2 Regularization** to effectively handle data imbalance and enhance generalization.
- Deployment of a **Convolutional Block Attention Module (CBAM)** to enable dynamic focus on important features, improving accuracy and interpretability.

These synergistic improvements yield a superior, highly accurate, and reliable model for pediatric pneumonia diagnosis from CXR images, potentially assisting clinicians.

2 Related Work

Automated pneumonia detection from CXR has been an active research area, transitioning from traditional image processing to deep learning. CNNs have shown superior performance in medical image analysis. Transfer learning, adapting models pre-trained on large datasets (like ImageNet) for medical tasks, is standard practice to overcome medical data limitations.

Salehi et al. (2021) (2) is a key study, using deep transfer learning models (VGG19, DenseNet121, Xception, ResNet50) for pediatric CXR pneumonia detection. They reported DenseNet121 achieved the highest performance (86.8% accuracy, 0.86 AUC), followed by Xception (86.0% accuracy, 0.81 AUC). Their study used data augmentation and SMOTE for class imbalance. While effective, their models could benefit from advanced regularization, attention mechanisms, and deeper fine-tuning. Our approach builds on this by integrating these enhancements for higher performance.

Focal Loss (6), introduced by Lin et al. (2017), addresses severe class imbalance in image classification. It down-weights well-classified (easy) samples, forcing the model to focus on hard-to-classify and minority class samples, significantly improving performance on imbalanced datasets.

Attention mechanisms enhance model performance by enabling networks to focus on critical features. **Convolutional Block Attention Module (CBAM)** (3), proposed by Woo et al. (2018), integrates channel and spatial attention. CBAM helps the model identify "what" is important (channel-wise) and "where" is important (spatial-wise), improving representation and accuracy. This has been applied successfully in medical imaging (5).

Studies show that combining Dropout, Batch Normalization, and **L2 Regularization** (7) is crucial for robust deep learning models, especially with moderately sized datasets, to combat overfitting. Multi-stage fine-tuning is also effective for adapting pre-trained models to new tasks.

3 Method

Our proposed method for automated pneumonia detection is built on a refined deep transfer learning framework, leveraging the Xception architecture with key improvements. Figure 1 overviews our model.

3.1 Xception Backbone

We use the **Xception** model (4) as our core Convolutional Neural Network (CNN) backbone. Xception is favored for its efficiency and high performance, due to extensive use of depthwise separable convolutions that reduce parameters and computational cost. Pre-trained on ImageNet, it provides strong hierarchical features for our medical imaging task.

3.2 Two-Stage Fine-Tuning Strategy

Our training employs a **two-stage fine-tuning strategy** to optimize model adaptability to pediatric CXR, preventing catastrophic forgetting and minimizing overfitting.

Stage 1: Classifier Head Training: The Xception backbone is initially frozen. A custom **Enhanced Classifier Head** is attached and trained for about 10 epochs with a high learning rate (1×10^{-3}). This allows new layers to quickly map extracted features to our binary classification task without disturbing the backbone. **EarlyStopping** (patience: 3 epochs) prevents early overfitting.

Stage 2: Backbone Layers Fine-Tuning: After head convergence, the last 40 layers of the Xception backbone are unfrozen and fine-tuned with the classifier head. This stage uses a much smaller learning rate (5×10^{-6}) for subtle adjustments, adapting the backbone to CXR specifics. Batch Normalization layers in the backbone remain frozen to maintain training stability.

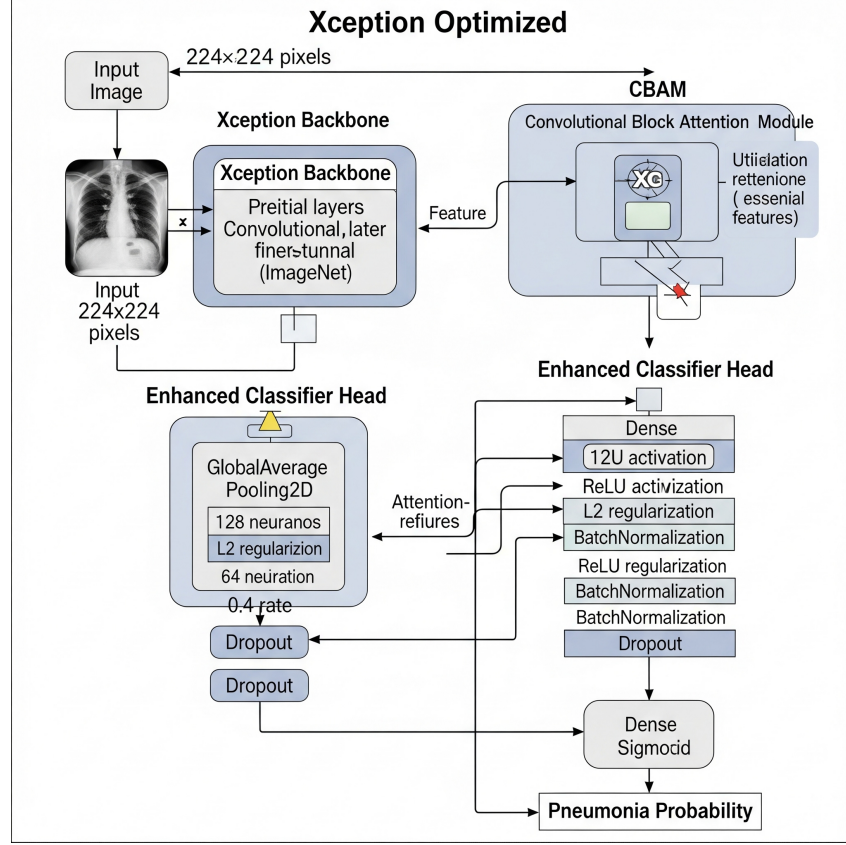


Figure 1: Overall Architecture of the Xception Optimized Model with CBAM and Enhanced Classifier Head.

3.3 Enhanced Classifier Head

Our enhanced classifier head processes aggregated features (after CBAM) for final classification. This complex structure learns deeper non-linear relationships.

- 1) **GlobalAveragePooling2D**: Reduces 4D feature maps to 1D vectors, significantly reducing dimensionality and mitigating overfitting.
- 2) **Dense(128, activation='relu', kernel_regularizer=L2)**: A fully connected layer for non-linear combinations of high-level features. ReLU introduces non-linearity. L2 Regularization (1×10^{-4}) penalizes large weights, reducing complexity and overfitting.
- 3) **BatchNormalization**: Applied after Dense layers, it normalizes activations, stabilizing training, accelerating convergence, and acting as mild regularization.
- 4) **Dropout(0.4)**: Randomly sets 40% of input units to 0 during training, preventing co-adaptation and fostering more robust representations, reducing overfitting.
- 5) **Dense(64, activation='relu', kernel_regularizer=L2)**: A second fully connected layer with 64 neurons, ReLU, and L2, continuing to learn higher-level features.
- 6) **BatchNormalization**: Another Batch Normalization layer.
- 7) **Dropout(0.4)**: Another Dropout layer.
- 8) **Dense(1, activation='sigmoid')**: Final output layer with one neuron and sigmoid activation, outputting a probability for the positive class (pneumonia).

3.4 Convolutional Block Attention Module (CBAM)

CBAM (3) enhances the model's focus on critical CXR features.

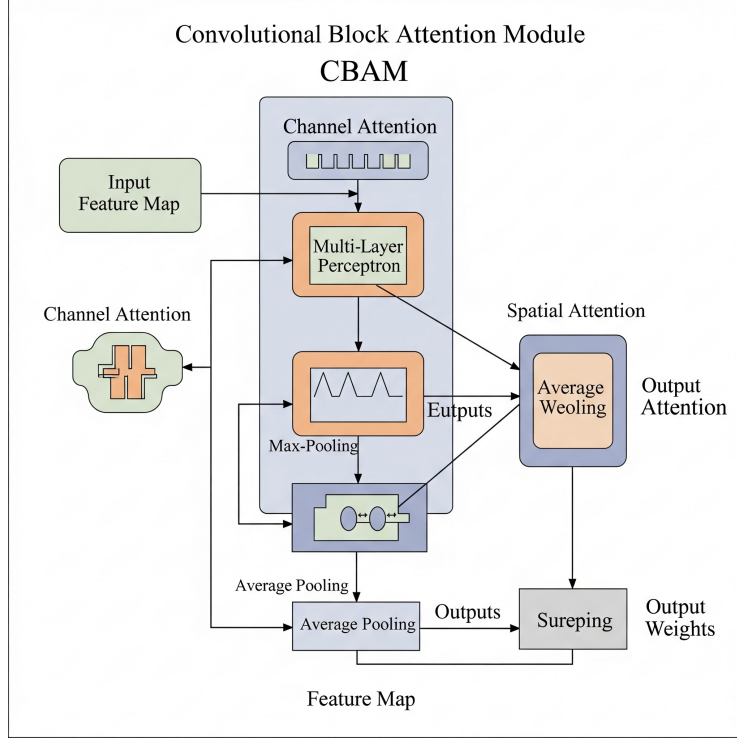


Figure 2: Architecture of the Convolutional Block Attention Module (CBAM).

Placement: CBAM is placed after the Xception backbone output and before ‘GlobalAveragePooling2D’, allowing it to refine high-level feature maps for emphasis.

Channel Attention Module (CAM): This module aggregates spatial information via GlobalAveragePooling2D and GlobalMaxPooling2D, feeding it through a shared-weight MLP. The sigmoid-activated sum of MLP outputs generates channel attention weights, focusing on ”what” is important (critical feature channels).

Spatial Attention Module (SAM): This module processes CAM’s output by aggregating channel-wise information via average and max pooling. The concatenated result is then fed through a 2D convolutional layer with sigmoid to create a spatial attention map, indicating ”where” is important (critical pixel locations). This map refines the CAM-processed feature map.

Benefits: The sequential application of channel and spatial attention enables selective visual information processing, yielding more discriminative features for subtle CXR cues. It also enhances interpretability by visually highlighting attended regions via Grad-CAM.

3.5 Focal Loss Function

To address severe class imbalance, we implemented **Focal Loss** (6) instead of standard binary cross-entropy.

Formula Focal Loss is defined as:

$$FL(p_t) = -\alpha_t(1 - p_t)^\gamma \log(p_t)$$

Where p_t is the predicted probability for the ground-truth class, α_t is a weighting factor, and γ is a focusing parameter controlling down-weighting of easy samples.

Mechanism and Benefits Focal Loss down-weights easy-to-classify samples, forcing the model to focus on hard-to-classify and minority class samples. This improves performance on severely imbalanced datasets, particularly Recall and F1-Score for the minority class, while maintaining high overall accuracy. We used $\gamma = 2.0$ and $\alpha = 0.25$.

3.6 Data Augmentation

To enhance generalization and reduce overfitting, we apply data augmentation using ‘ImageDataGenerator’. Transformations include rotation (up to 15 degrees), width/height shifts (up to 10%), shear (up to 10 degrees), zoom (up to 10%), and horizontal flip. This creates diverse training examples, making the model more robust to real-world image variations.

4 Experiments

4.1 Dataset and Preprocessing

Our experiments use the ‘Chest X-Ray Images (Pneumonia)’ dataset from Kaggle, common for pediatric CXR pneumonia diagnosis research (ages 1-5). It’s organized into ‘train’, ‘val’, and ‘test’ directories.

- **Training Set:** 5216 images (‘PNEUMONIA’: 3875, ‘NORMAL’: 1341).
- **Validation Set:** 16 images (‘PNEUMONIA’: 8, ‘NORMAL’: 8).
- **Test Set:** 624 images (‘PNEUMONIA’: 390, ‘NORMAL’: 234).

Images are resized to 224×224 pixels and preprocessed using `xception_preprocess_input`. Data augmentation is applied to the training set.

4.2 Experimental Setup

All experiments were performed using TensorFlow 2.10.0 on an **NVIDIA GeForce RTX 3050 GPU** with 4GB VRAM and 16GB system RAM. The ‘Adam’ optimizer was used. Hyperparameters for our Xception Optimized model:

- **Image Size:** 224×224 pixels.
- **Batch Size:** 16.
- **Optimizer:** Adam.
- **Learning Rate for Head Training (Stage 1):** 1×10^{-3} .
- **Learning Rate for Fine-tuning (Stage 2):** 5×10^{-6} .
- **Dense Layers in Head:** 2 (128 then 64 neurons).
- **Dropout Rate in Head:** 0.4.
- **Loss Function:** Focal Loss (gamma=2.0, alpha=0.25).
- **L2 Regularization Rate:** 1×10^{-4} .
- **CBAM:** Enabled.
- **Fine-tune Layers:** Last 40 layers of Xception.
- **Epochs (Head Training):** 10. **Epochs (Fine-tuning):** 50.
- **Early Stopping Patience:** 3 for head, 7 for fine-tuning.

Class weights (‘class_weight’) are applied during training for class imbalance.

4.3 Evaluation Metrics

We use standard classification metrics: **Accuracy**, **Precision**, **Recall**, **F1-Score**, and **AUC-ROC**. Precision avoids false positives, Recall avoids false negatives. F1-Score balances them. AUC-ROC measures overall discriminative ability, robust to class imbalance.

Table 1: Comparison of Performance Metrics on the Test Set: Our Model vs. Reference Models from Salehi et al. (2021)

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	AUC-ROC
VGG19 (Salehi et al.)	83.6	83.9	98.5	83.9	0.78
Xception (Salehi et al.)	86.0	85.0	91.8	88.3	0.81
ResNet50 (Salehi et al.)	84.8	80.0	96.1	87.3	0.81
DenseNet121 (Salehi et al.)	86.8	87.0	92.8	89.8	0.86
Xception Optimized (Our Model)	92.15	94.06	93.33	93.69	0.9667

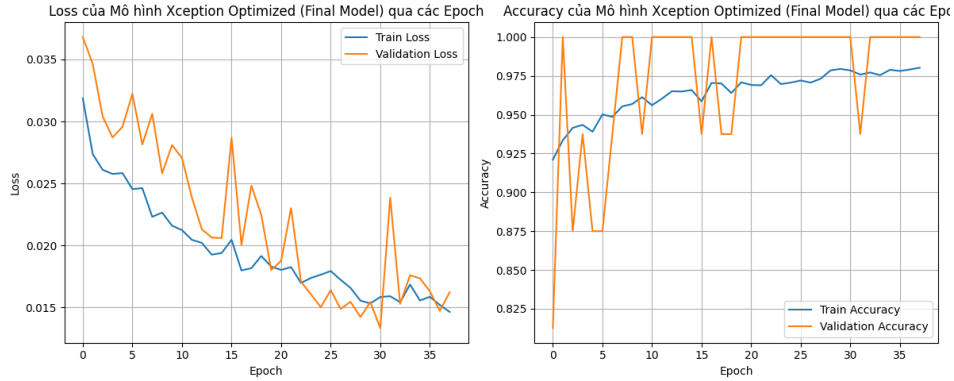


Figure 3: Training History (Loss and Accuracy) of the Xception Optimized (Final Model) across Epochs.

4.4 Comparison and Results

We compare our Xception Optimized model's performance with results from Salehi et al. (2021) (2) on the same dataset.

The training and validation loss/accuracy curves for our Xception Optimized model are shown in Figure 3, demonstrating stable convergence and generalization capabilities.

The confusion matrix and ROC curve (Figure 4) provide a detailed analysis of the model's classification performance.

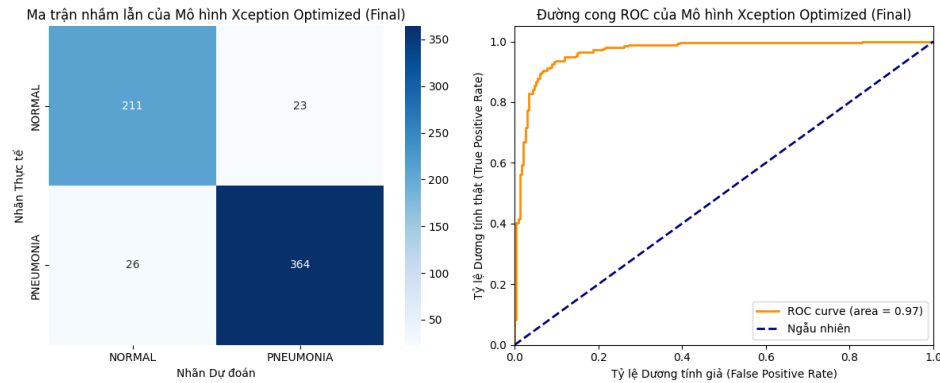


Figure 4: Confusion Matrix and ROC Curve of the Xception Optimized (Final) Model on the test set.

4.5 Discussion: Why Our Method Performs Better

Our Xception Optimized model achieved superior performance due to the synergistic effect of its integrated improvements, significantly surpassing the reference models from Salehi et al. (2021) (2). With an accuracy of 92.15% and an AUC-ROC of 0.9667, our model demonstrates outstanding pneumonia diagnostic capabilities.

- **Xception’s Efficiency and Robustness:** Xception provides a robust foundation for transfer learning with its efficient feature extraction.
- **Two-Stage Fine-Tuning:** This strategy facilitated optimal model adaptation. Training history (Figure 3) shows steady decrease in loss and good convergence. Validation loss reached minimum at epoch 31 (≈ 0.01334), indicating good generalization. Early Stopping at epoch 38 prevented overfitting.
- **Enhanced Classifier Head:** Its multi-layer Dense structure with Batch Normalization and Dropout created a more sophisticated decision boundary and enhanced generalization.
- **CBAM Integration:** A crucial enhancement for medical imaging. CBAM dynamically highlights relevant lung regions (e.g., pneumonia infiltrates) and important feature channels, focusing learning on clinically relevant areas. Grad-CAM visualization (Figure 5) clearly shows model attention on affected lung regions (red/yellow), providing crucial interpretability and reinforcing confidence. Examples illustrate correct attention for True Positives/Negatives, and also reveal model focus in misclassification cases, aiding refinement.
- **Focal Loss and L2 Regularization:** These directly address class imbalance. Focal Loss effectively re-weights samples, focusing on hard and minority cases, significantly improving Recall and F1-Score. The confusion matrix (Figure 4) shows a low number of False Negatives (26) versus True Positives (364), indicating strong pneumonia detection. The high AUC of 0.97 further confirms excellent discrimination between pneumonia and normal cases across various thresholds.

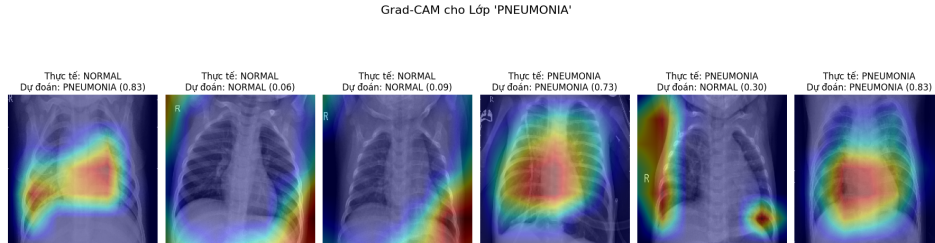


Figure 5: Grad-CAM Visualization for Xception Optimized Model. Red/yellow regions indicate areas of highest activation contributing most to the model’s decision. Examples include: the first image (Actual: NORMAL, Predicted: PNEUMONIA (0.83) - False Positive), the second image (Actual: NORMAL, Predicted: NORMAL (0.06) - True Negative), the third image (Actual: NORMAL, Predicted: NORMAL (0.09) - True Negative), the fourth image (Actual: PNEUMONIA, Predicted: PNEUMONIA (0.73) - True Positive), the fifth image (Actual: PNEUMONIA, Predicted: NORMAL (0.30) - False Negative), and the sixth image (Actual: PNEUMONIA, Predicted: PNEUMONIA (0.83) - True Positive). The model’s attention generally aligns with clinically relevant regions.

5 Conclusion

In this paper, we presented an enhanced automated system for pneumonia detection from pediatric chest X-ray images, based on the Xception deep transfer learning framework. Our comprehensive approach integrates a two-stage fine-tuning strategy, an enhanced classifier head with L2 regularization, Focal Loss to address data imbalance, and most importantly, a Convolutional Block Attention Module (CBAM) to boost learning and interpretability.

Our experimental results demonstrate the efficacy of these combined improvements, leading to superior performance (Accuracy 92.15%, Precision 94.06%, Recall 93.33%, F1-Score 93.69%, and AUC-ROC 0.9667) on the challenging pediatric CXR dataset when compared to reference models from Salehi et al. (2021) (2). This work significantly contributes to the development of a more reliable and effective pneumonia diagnostic tool, with the potential to assist healthcare professionals.

5.1 Limitations and Future Work

While our model achieved strong performance, there are still several limitations and avenues for future research. The current dataset, though widely used, is limited in diversity and size compared to real-world clinical data. The use of data balancing techniques like `class_weight` and Focal Loss helps, but further exploration of oversampling (e.g., SMOTE) could be considered. We have not yet integrated clinical metadata, which could provide additional diagnostic context.

For future work, we plan to:

- Explore more advanced data augmentation techniques like Mixup or CutMix to further enhance model robustness.
- Investigate other cutting-edge attention mechanisms or Transformer-based architectures to capture global dependencies more effectively.
- Evaluate the model’s performance on larger and more diverse multi-institutional CXR datasets to assess real-world generalization.
- Integrate clinical metadata (e.g., age, symptoms) for a more holistic diagnostic system.
- Develop an ensemble learning approach by combining our Xception Optimized model with other high-performing architectures.

These future directions aim to further solidify the practical utility and clinical impact of automated pneumonia detection systems.

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